

Advanced usage

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About advanced usage

Some of the operations and functions introduced in this chapter do not normally need to be used. We recommend that you do not use it if you do not understand the term or meaning.

Meg

I will explain how to use Meg, which was not mentioned in the basic usage.

Unconfirmed transmission

[communication]

By exchanging WonderSwan with each other, it will be sent without displaying the confirmation dialog on the receiving side.

1. Connect WonderSwan to each other with a communication battle cable.
2. Make Meg of both WonderSwan the communication screen.
3. Display the menu on the sending side.
4. Select <Unconfirmed Send> with the X button.
5. Press the A button to execute.
6. The file will be transferred.

* Please note that if you turn off the power during transmission, the data will be destroyed.

* If you interrupt during transmission / reception, a garbage file may be generated. Garbage files are the same as deleted files. Until the area is tidied up, the free space will be reduced by the size of the garbage file.

XMODEM reception

[communication]

Use XMODEM to receive WonderWitch transfer format files.

Use when transferring files directly with your favorite XMODEM tool without using TransMagic.

1. PC-Connect your PC to WonderSwan with a WonderSwan connection cable.
 2. Make Meg the communication screen.
 3. Send the WonderWitch transfer format file with your favorite XMODEM tool.
 4. Show menu
 5. Select <Receive XM> with the X button
 6. Press the A button to execute.
 7. The file will be transferred.
-

XMODEM transmission

[communication]

Convert the file on the cartridge to WonderWitch transfer format and send the file with XMODEM.

Use when transferring files directly with your favorite XMODEM tool without using TransMagic.

1. Connect your PC to WonderSwan with a PC-WonderSwan connection cable.
 2. Make Meg the communication screen.
 3. Select the file to send.
 4. Show menu
 5. Select <Send XM> with the X button
 6. Press the A button to execute.
 7. Receive with your favorite XMODEM tool on the PC side.
 8. The file will be transferred.
-

Remote shell

[communication]

You can operate Meg via the serial port on the communication screen.

This interface for operating from a PC is called a "remote shell" in WonderWitch.

See the appendix "Remote Shell Protocol Specification" for the commands available in Remote Shell and their formats.

Main commands

```
ls df put get delete exec HELO stty reboot newfs defrag rename copy speed date setinfo chmod cd  
pwd
```

In order to use the remote shell, it is necessary to start the terminal software such as Tera Term Pro on the PC side. (Be careful of the serial port conflict with TransMagic)

Set the terminal software as follows.

```
speed: 38400bps  
data: 8bit
```

```
parity: none
stop bit: 1
flow control: none
new-line: send CR / recv CR
kanji code: ShiftJIS
local echo: off
```

Immediately after switching to the communication screen, no prompt is displayed (non-interactive mode). You will be prompted by placing a line break only or issuing the "stty echo" command (interactive mode).

Enter the command after the prompt.

Simple mode startup

Simple mode is mainly used for recovery in case of problems.

When started in simple mode, the initialization file is not read, switched to the user shell, and restarted by suspend data is not performed.

1. Turn on the power switch while pressing the Y1 and Y4 buttons at the same time.
-

Startup settings

If you place the initialization file with the file name "init.rc" in the program area, its contents will be reflected at startup.

The following environment constants can now be defined as settings related to Meg.

```
$ MEG_KEYBIND = <keymap number> Change button assignment
$ MEG_AUTORUN = <exec parameter> Specify autorun program
```

The format is as follows.

```
$ Environment constant name = value <newline>
```

* There must be no space before or after the '='.

* For details on the initialization file, refer to "Software Architecture" in the Developer's Manual.

\$ MEG_KEYBIND Button assignment change

```
$ MEG_KEYBIND = 0
```

```
A: Enter / Execute
B: Cancel / Menu ON / OFF
X: Select
Y1, 3: Menu ON / OFF
Y2: Screen switching Clockwise
Y4: Screen switching Counterclockwise
START: Pause / Resume
SOUND: Volume control
```

```
$ MEG_KEYBIND = 1
```

```
A: Enter / Execute
B: Cancel / Switch screen Counterclockwise
X: Select
Y1,2,3: Menu ON / OFF
START: Pause / Resume
SOUND: Volume control
```

```
$ MEG_KEYBIND = 2
```

```
A: Enter / Execute  
B: Cancel / Menu ON / OFF  
X: Select  
Y2,3: Screen switching Clockwise  
Y1,4: Screen switching Counterclockwise  
START: Pause / Resume  
SOUND: Volume control
```

```
$ MEG_KEYBIND = 3
```

```
A: Enter / Execute  
B: Cancel / Menu ON / OFF  
X: Select  
Y1, 3: Menu ON / Select  
Y2: Screen switching Clockwise  
Y4: Screen switching Counterclockwise  
START: Pause / Resume  
SOUND: Volume control
```

\$ MEG_AUTORUN Specifying an autorun program

(Example) The image file is displayed for 500 unit hours at startup.

```
$ MEG_AUTORUN = / rom0 / imager imagefile 500
```

Shell resume suppression function when the power is turned on

If you turn off the power and exit WonderWitch while the application is running, the shell will resume the next time you turn it on. This is because when executing a user program, the shell suspends itself and then starts the user program, but when specifying an autorun program by \$ MEG_AUTORUN, etc., it is not his resume at power cycle, but the initial time. You may want to boot from the state.

For such cases, the function that suppresses shell resume when FreyaOS starts is the "shell resume suppression function". To use this function, set the following environment constants in the initialization file /rom0/init.rc.

```
$ DISABLE_SHELL_RESUME = true
```

For example, even if the calendar program (/ rom0 / calendar) is executed when the power is turned on and the power is turned off while the calendar program is being executed, in order to start the calendar program again when the power is turned on again, a file describing the following contents is described. In /rom0/init.rc.

```
$ MEG_AUTORUN = / rom0 / calendar  
$ DISABLE_SHELL_RESUME = true
```

* This function can be used with FreyaOS 1.0.3 or later versions.

TransMagic

I will explain how to use TransMagic, which was not touched on in the basic usage.

Display of cartridge information

1. Connect TransMagic.
2. Execute Cartridge Menu → Cartridge Information.
3. Cartridge information is displayed. (Press OK to turn it off)

Area tidy

Organizes the selected file area.

1. Connect TransMagic.
 2. Select the file area you want to organize.
 3. Execute Cartridge Menu → Organize Area.
 4. Area tidying is done.
-

Clear all areas

Initializes the selected file area.

1. Connect TransMagic.
 2. Select the file area you want to clear.
 3. Cartridge menu → Clear all areas.
 4. All areas will be cleared.
-

Serial port settings

Set the serial port of the PC. (No settings are made on the WonderSwan side.)

1. **Disconnect** TransMagic .
 2. Select File menu → Serial port settings.
 3. In the dialog, select the communication port and communication speed to use correctly.
 4. Press OK to set.
-

Clear protocol log

Clear the protocol log (recording of communication contents) with WonderSwan.

1. Select File-> Clear Protocol Log
 2. The protocol log display area is cleared
-

Saving protocol logs

Save the protocol log with a file name.

1. Select File-> Save Protocol Log
2. A dialog for entering the name of the file to save is displayed.

Enter the file name and press OK.

Monitor mode

It is not used for anything other than OS upgrade / backup.

If you turn on the power switch while pressing the B button, the monitor mode will be activated with or without suspend information.

Start monitor mode

Hold down the B button and turn on the power switch.

Enter monitor mode. Monitor mode is used when upgrading / backing up the OS and setting the cartridge.

Now, if you hold down the A + B button and turn on the power switch, the VT100 escape sequence is also used to display the monitor screen on the serial port. It can be operated using the keyboard of the PC.

Monitor mode menu

In monitor mode, you can upgrade the OS, set the time, set the serial port speed and sleep time.

Monitor mode menu

Setup Displaying the cartridge setup menu

Recv System OS Upgrade

Send System OS Backup

Recv Soft Not

Used Send Soft Not Used

Exit Exit monitor mode

Setup menu

Date Date

Time Time

Baudrate Serial Port Speed

Sleep LCD Time to sleep

Using the user shell

FreyaOS will use a file named "@shell" in the program area (/ rom0) as a user-specified shell. However, this file must be an indirect library for RunnableIL.

This mechanism allows users to use their preferred shell and developers to provide their preferred shell.

Even if you specify a user shell, pressing the Y1 and Y4 buttons at the same time launches the default shell (Meg). If you want to delete the user shell, start the default shell and then delete the "@shell" file in the program area.

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